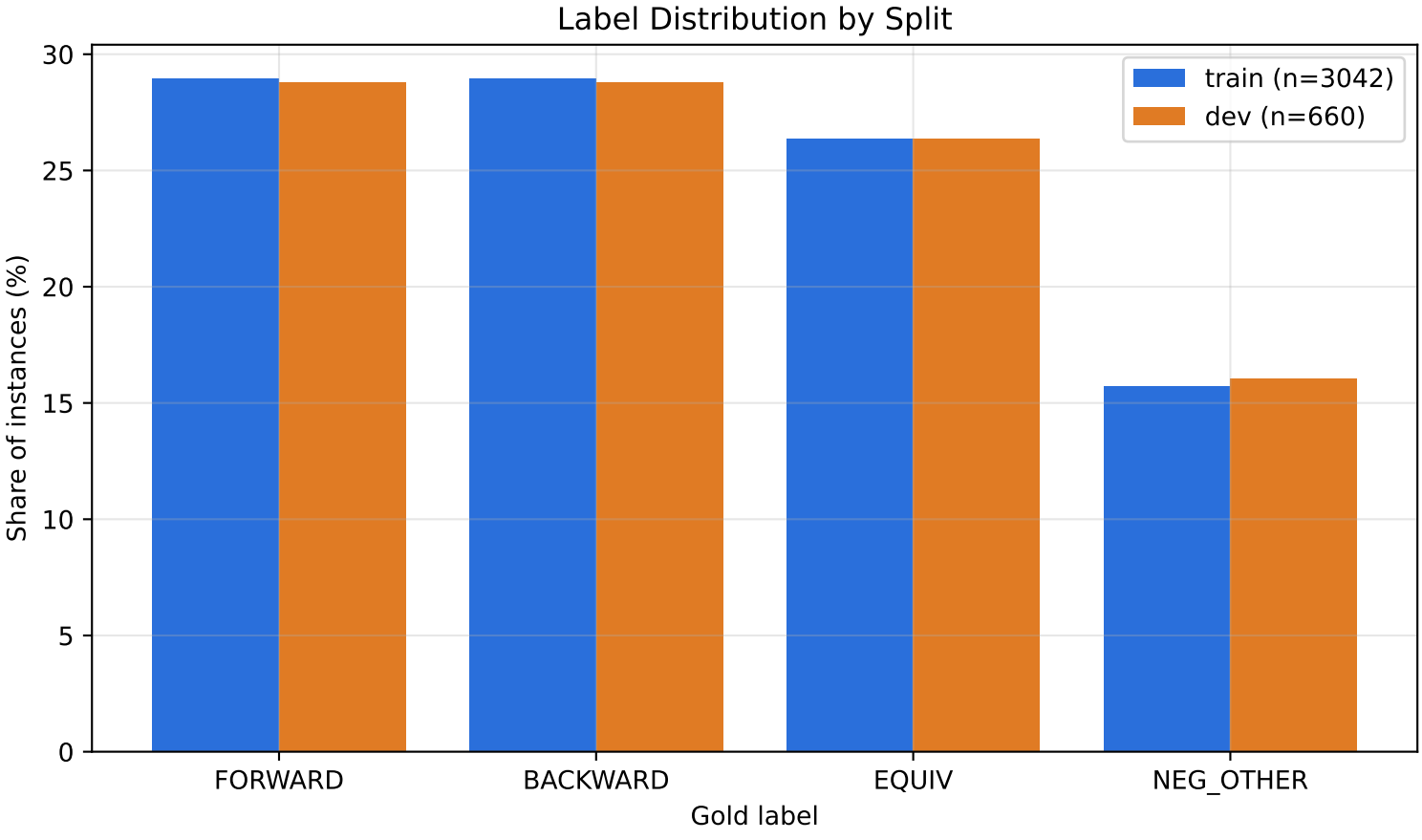
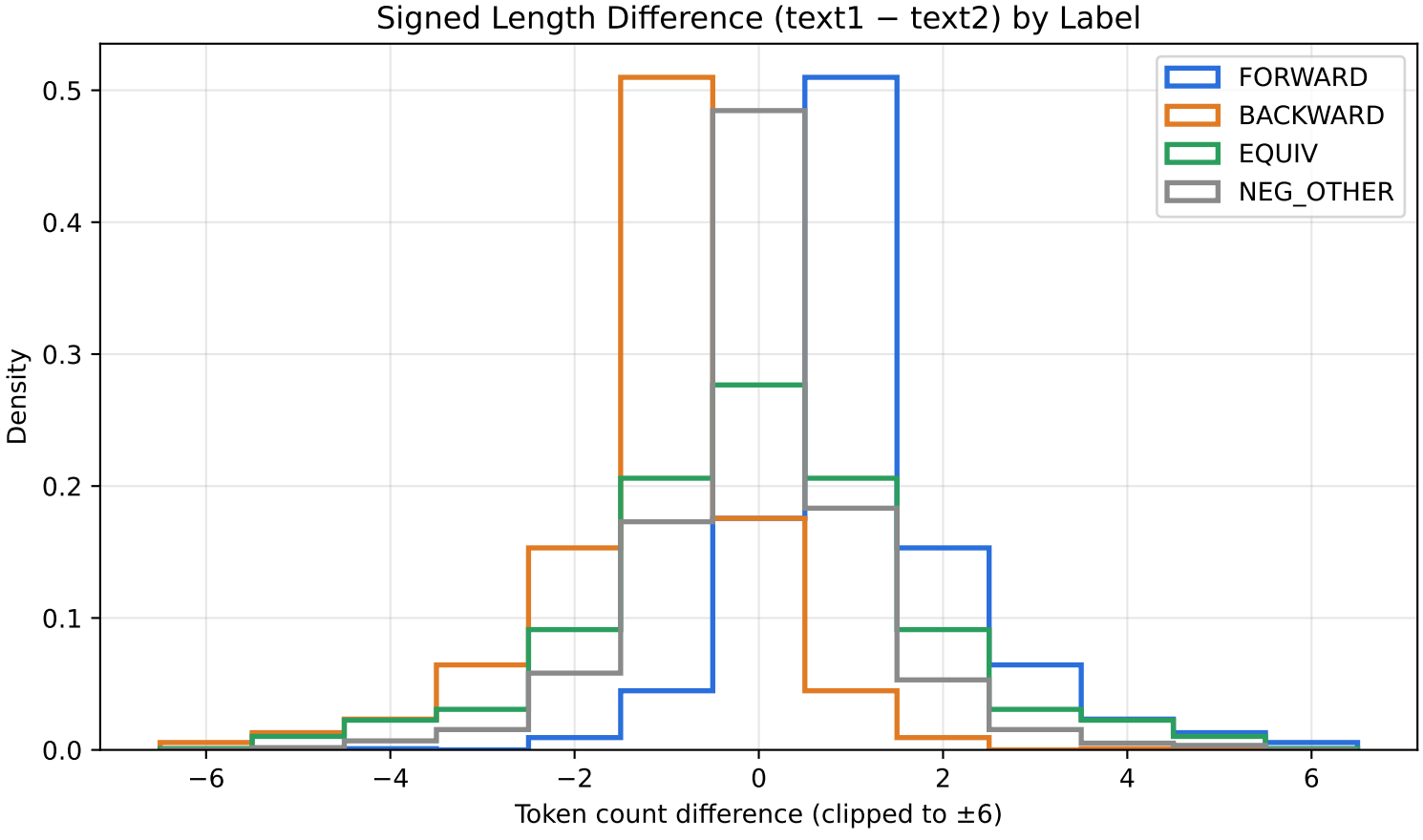
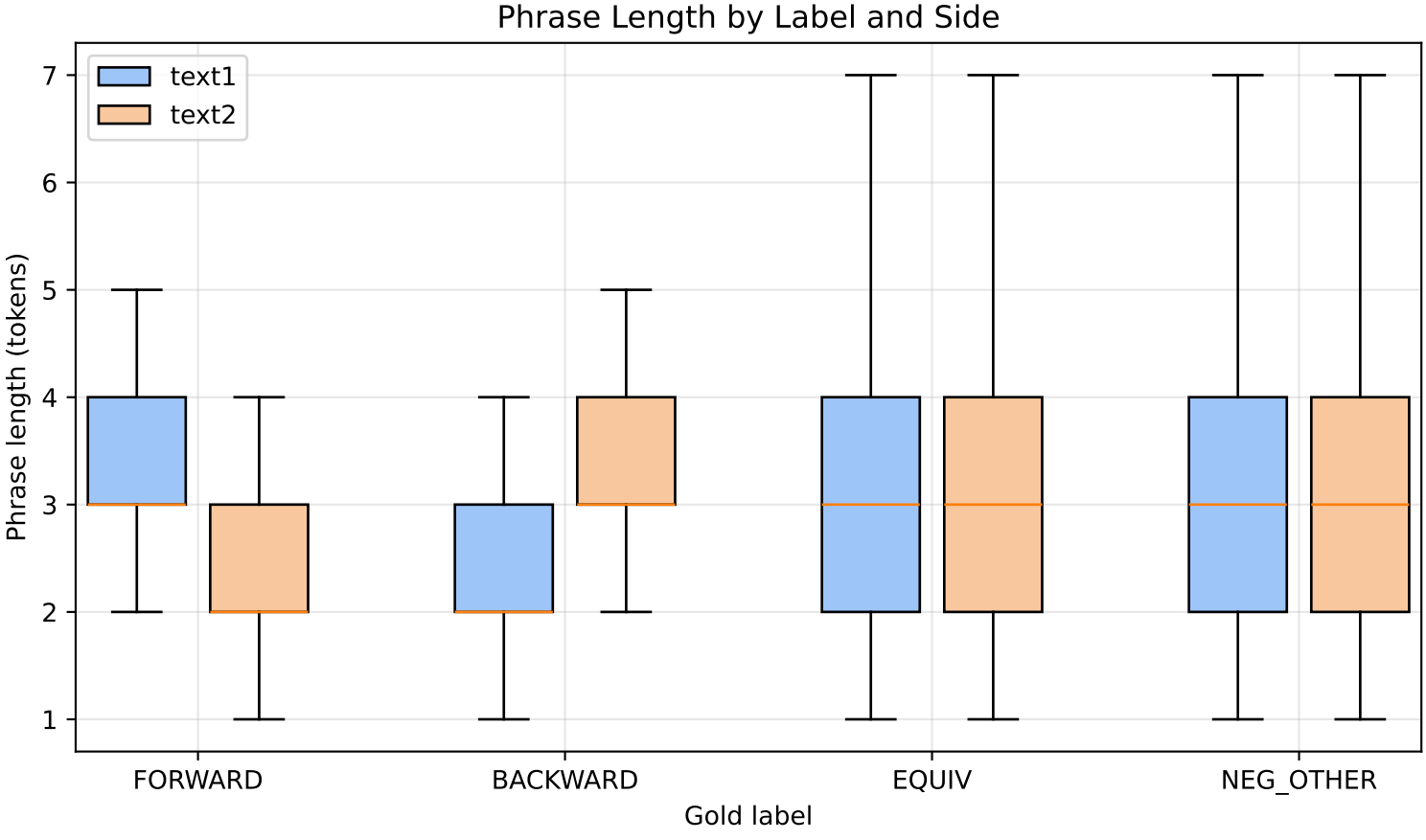
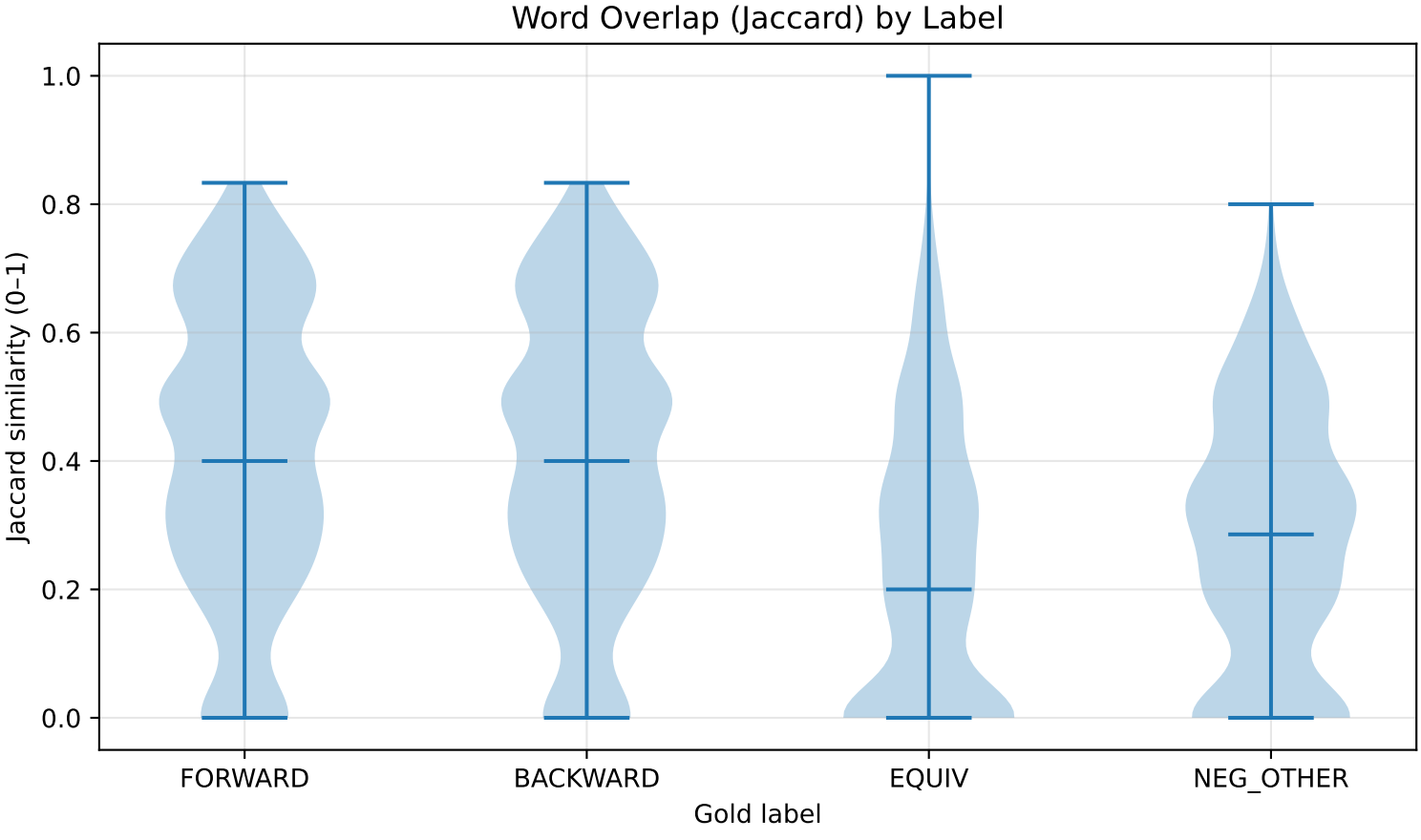
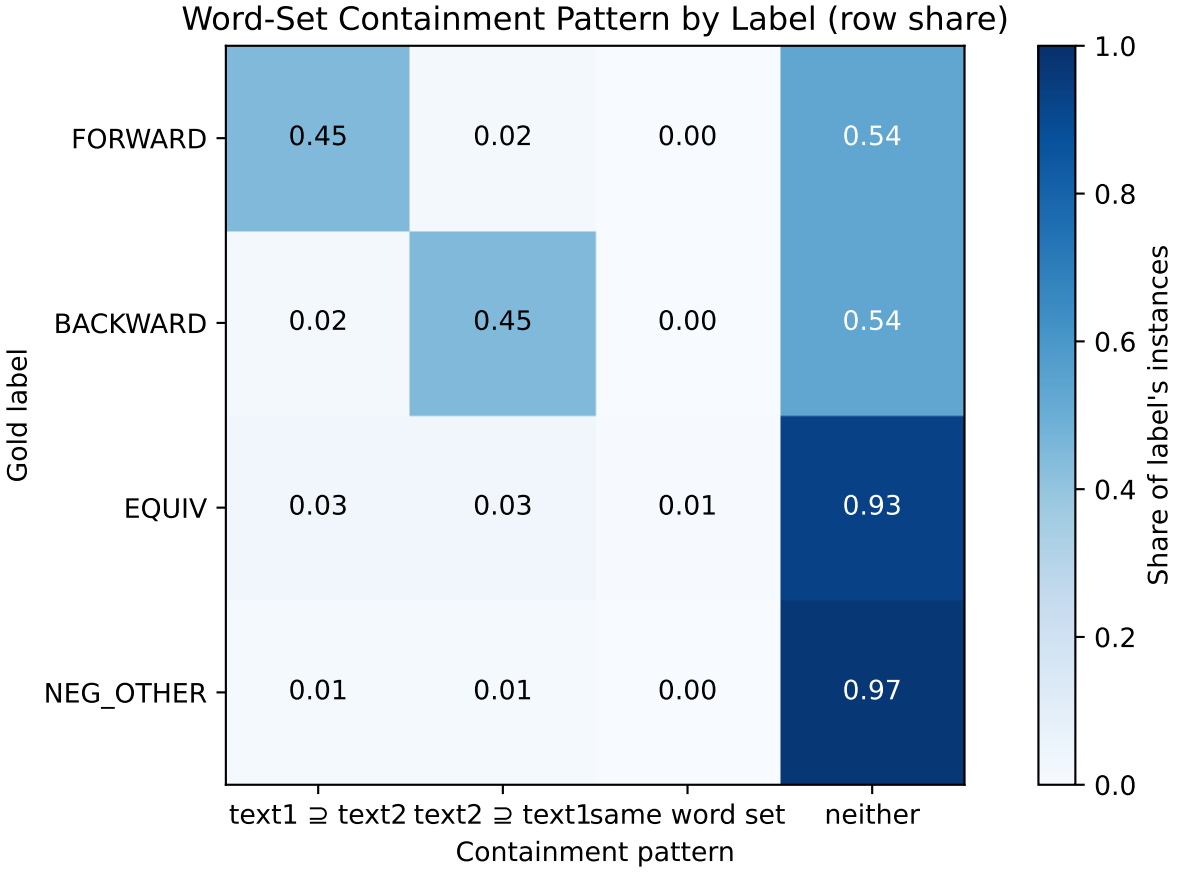
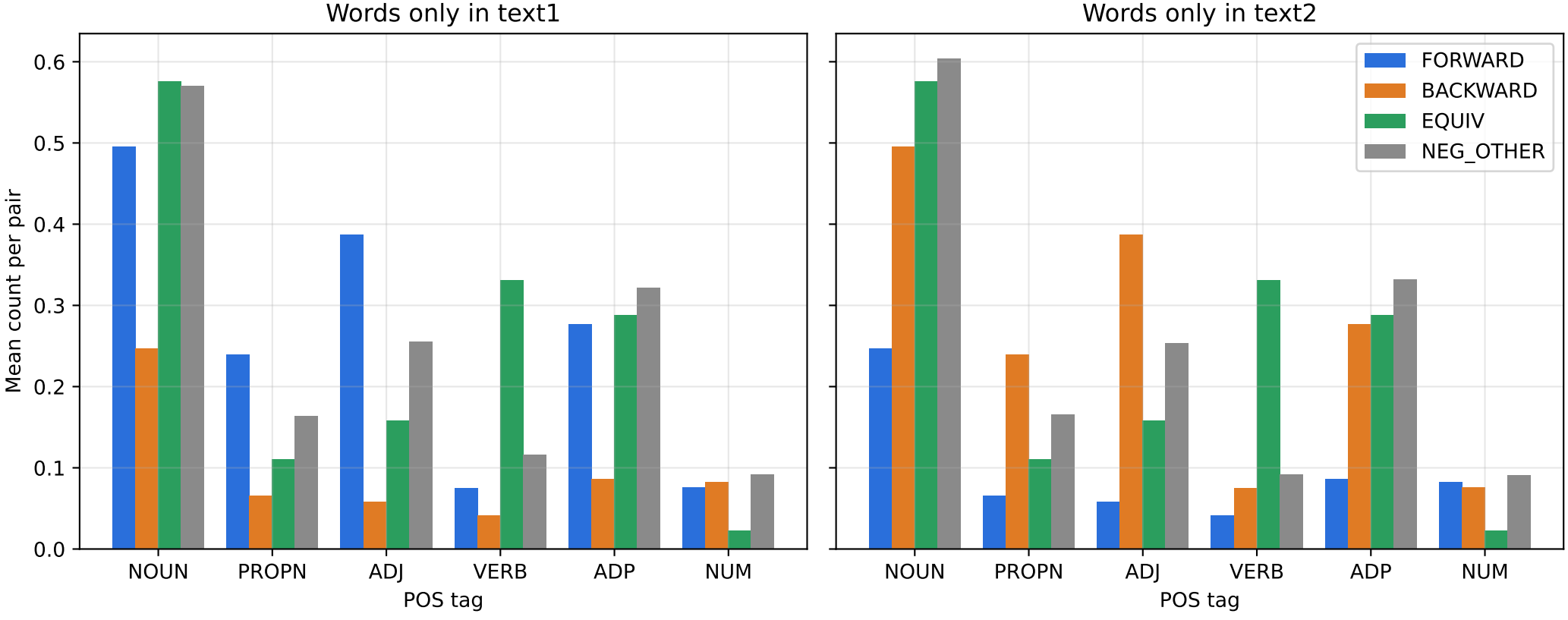

Label Distribution by Split
train (n=3042)
dev (n=660)
Share of instances (%)
30
25
20
15
10
5
0
FORWARD
BACKWARD
EQUIV
NEG_OTHER
Gold label

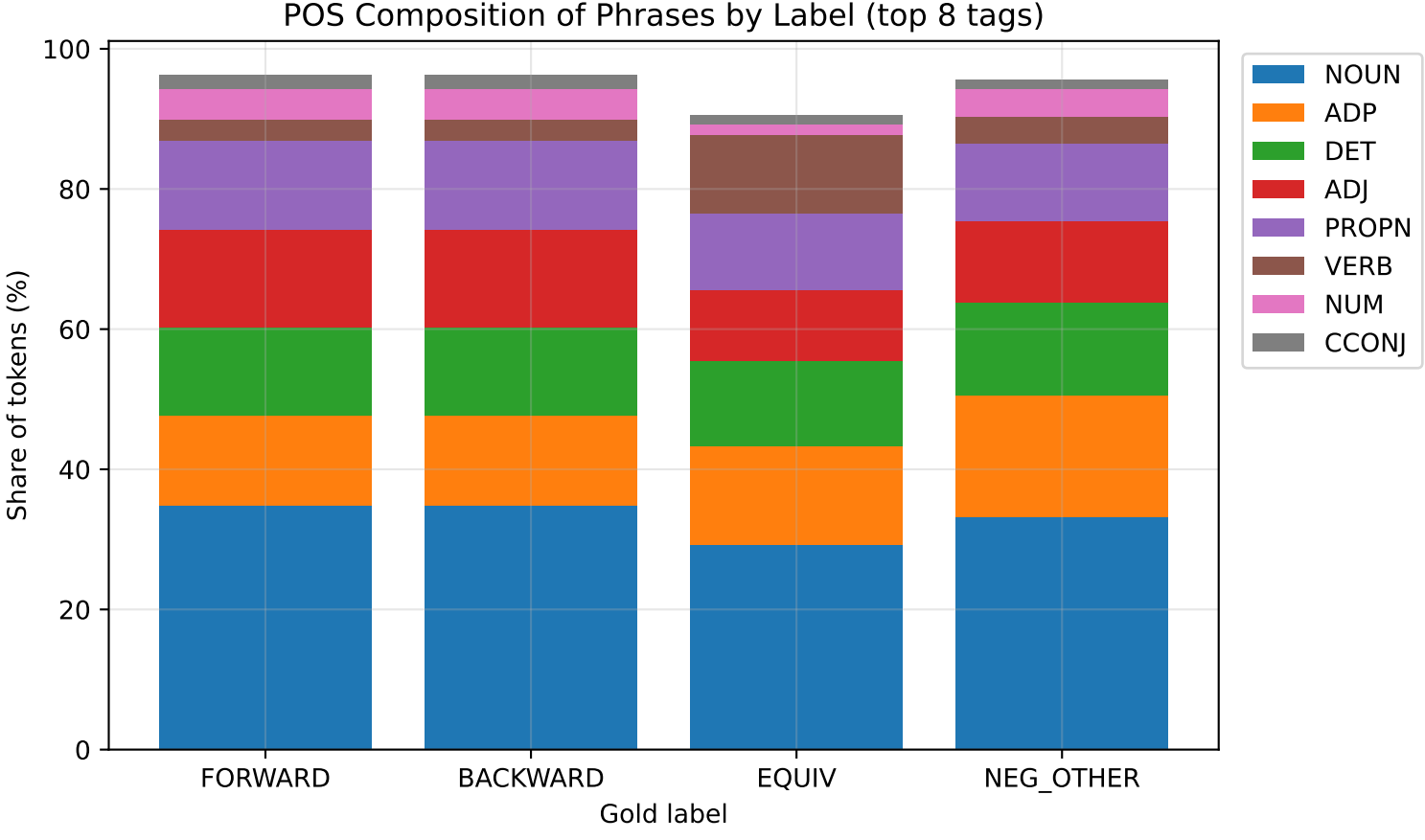

Signed Length Difference (text1 − text2) by Label
FORWARD
BACKWARD
EQUIV
NEG_OTHER
Density
0.0
0.1
0.2
0.3
0.4
0.5
−6
−4
−2
0
2
4
6
Token count difference (clipped to ±6)

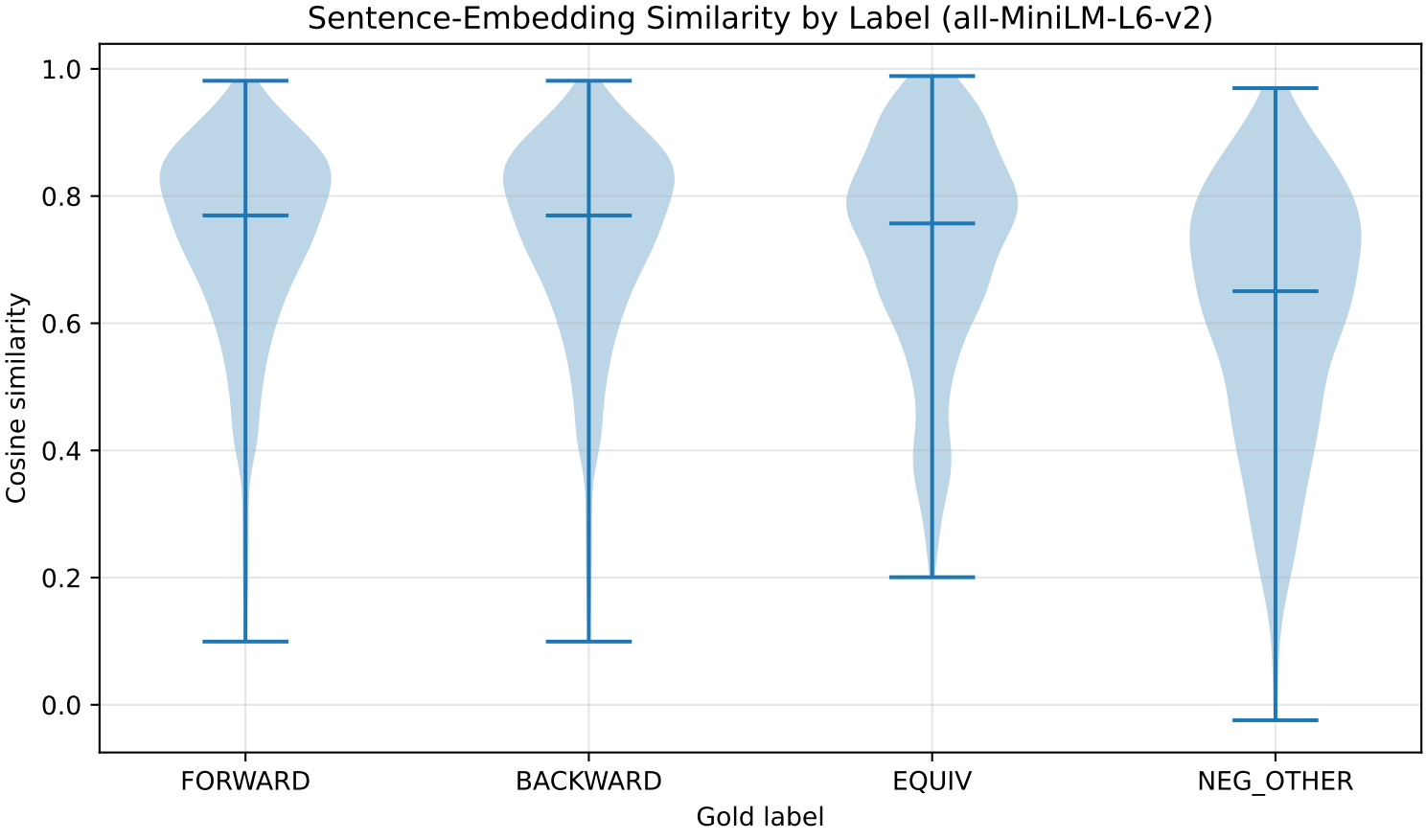

Phrase Length by Label and Side
text1
text2
Phrase length (tokens)
1
2
3
4
5
6
7
FORWARD
BACKWARD
EQUIV
NEG_OTHER
Gold label

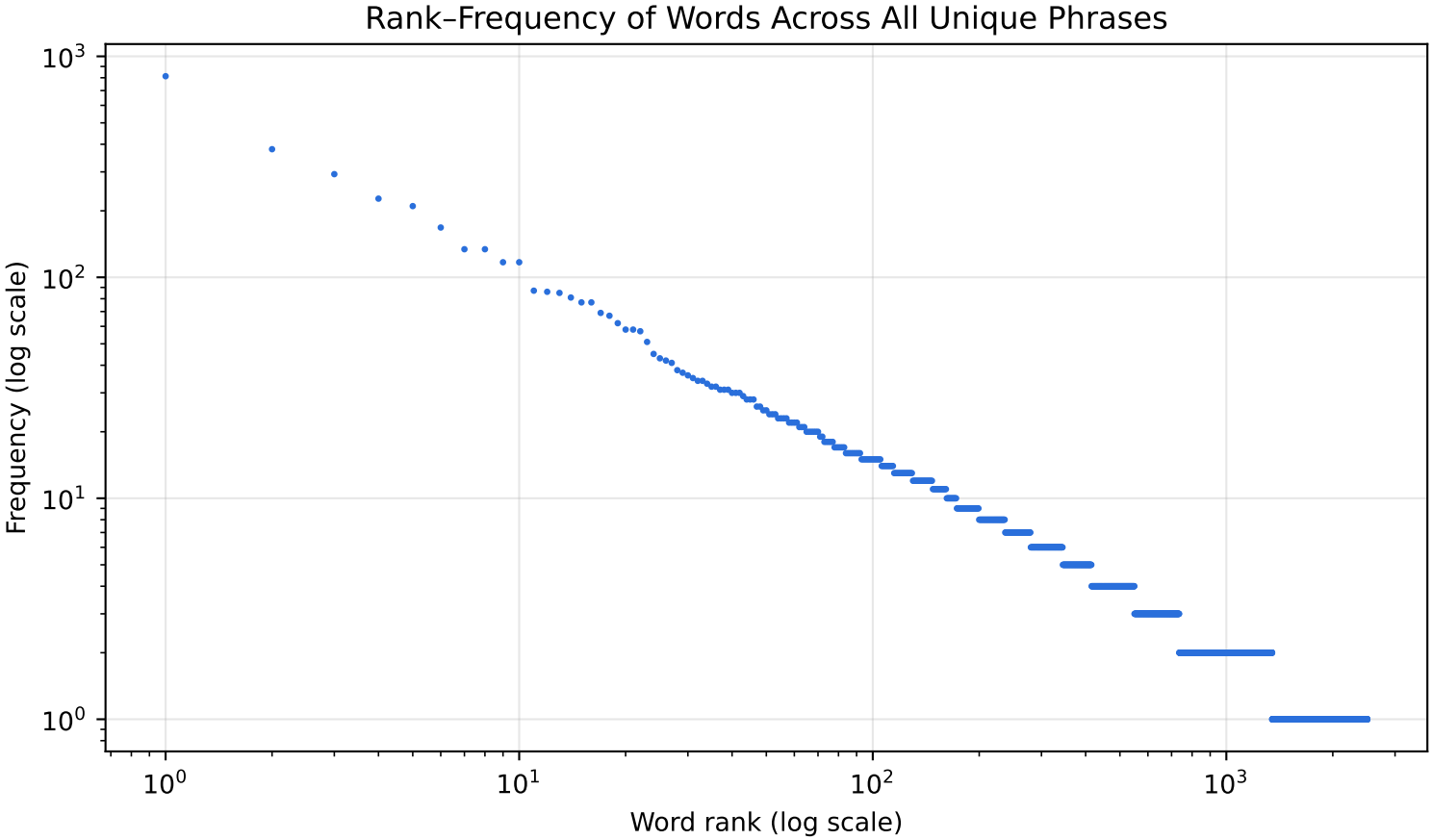

Word Overlap (Jaccard) by Label
Jaccard similarity (0–1)
1.0
0.8
0.6
0.4
0.2
0.0
FORWARD
BACKWARD
EQUIV
NEG_OTHER
Gold label


Word-Set Containment Pattern by Label (row share)
FORWARD
BACKWARD
EQUIV
NEG_OTHER
Gold label
0.45
0.02
0.00
0.54
0.02
0.45
0.00
0.54
0.03
0.03
0.01
0.93
0.01
0.01
0.00
0.97
text1 ⊇ text2
text2 ⊇ text1
same word set
neither
Containment pattern
Share of label's instances
1.0
0.8
0.6
0.4
0.2
0.0


POS of Non-Shared Words by Label
Words only in text1
Words only in text2
Mean count per pair
0.0
0.1
0.2
0.3
0.4
0.5
0.6
NOUN
PROPN
ADJ
VERB
ADP
NUM
POS tag
FORWARD
BACKWARD
EQUIV
NEG_OTHER


POS Composition of Phrases by Label (top 8 tags)
Share of tokens (%)
0
20
40
60
80
100
FORWARD
BACKWARD
EQUIV
NEG_OTHER
Gold label
NOUN
ADP
DET
ADJ
PROPN
VERB
NUM
CCONJ


Sentence-Embedding Similarity by Label (all-MiniLM-L6-v2)
Cosine similarity
1.0
0.8
0.6
0.4
0.2
0.0
FORWARD
BACKWARD
EQUIV
NEG_OTHER
Gold label


Rank–Frequency of Words Across All Unique Phrases
Frequency (log scale)
Word rank (log scale)
10^0
10^1
10^2
10^3